Finiteness of Shallow Logical Operators and a Sharp Depth Threshold

The Story Is Still Unfolding*
Yau Mathematical Sciences Center, Tsinghua University, Beijing 100084, China
Qiuzhen College, Tsinghua University, Beijing 100084, China
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For a quantum code on n finite-dimensional physical subsystems with distance d, we consider all exact logical unitaries realized by circuits of depth at most h and circuit gates acting on at most ρ subsystems. For a uniform upper-bound $h < 2\lceil \log_\rho d \rceil - 1$, these circuits can realize only finitely many logical actions up to a global phase, without any stabilizer or geometric locality assumptions. In contrast, for every qubit stabilizer code encoding at least one qubit, a non-trivial continuous one-parameter family of logical unitaries is available at depth $2\lceil \log_\rho d \rceil - 1$.

Controlling the error propagation is the central question in fault-tolerant computation. A single fault in a depth-h circuit with gate range ρ can affect at most ρ^h subsystems. So shallow quantum circuits of bounded range can effectively limit error propagation between error-correction steps. Under the constant-depth and bounded-range premise, Gottesman's threshold construction protects polynomial-size computations below a noise threshold with constant space overhead [1]. Nevertheless, the constant-depth premise seems not sufficient for complete, or even infinitely many logical operations. The celebrated Eastin–Knill theorem [2] shows that transversal gates with respect to a fixed partition into detectable blocks realize only finitely many logical actions up to a global phase. The same restriction holds for constant-depth circuits with geometric locality or algebraic assumptions [3–6]. Therefore, one may ask at what depth this necessary condition can first hold.

As we will show in the following, the precise threshold depth is $2\lceil \log_\rho d \rceil - 1$, given $d, \rho \geq 2$. More specifically, Theorem 1 rules out an infinite logical image when $h < 2\lceil \log_\rho d \rceil - 1$, without any stabilizer or geometric locality assumption. And Theorem 2 shows that every qubit stabilizer code encoding at least one qubit realizes a nontrivial continuous logical family at depth $2\lceil \log_\rho d \rceil - 1$, when arbitrary gates on at most ρ qubits are allowed. Stabilizer subspace codes are thus the most capable candidates for the emergence of an infinite logical family, attaining the smallest possible depth at given d and ρ.

Settings and notations. Let the physical system be $\mathcal{H} = \bigotimes_{i=1}^{n} \mathcal{H}_i$, with $n \geq 1$ and finite-dimensional factors $\mathcal{H}_i$. The code $\mathcal{C}$ is specified by an isometry $V : \mathcal{H}_L \otimes \mathcal{H}_G \longrightarrow \mathcal{H}$, where $\dim \mathcal{H}_L = K \geq 2$ and $\dim \mathcal{H}_G = g \geq 1$ denote the dimensions of logical and gauge subspaces, respectively. A distance lower bound d means that every operator O supported on fewer than d physical subsystems satisfies

$$V^\dagger O V = I_L \otimes M_O \tag{1}$$

for some operator M_O on $\mathcal{H}_G$ [7, 8]. A physical unitary U implements an exact logical action when

$$UV = V(W \otimes Z) \tag{2}$$

for unitaries W on $\mathcal{H}_L$ and Z on $\mathcal{H}_G$. This equality holds for all logical and gauge states, and the logical action is the phase class $[W] \in \mathrm{PU}(K)$. For integers $h \geq 1$ and $\rho \geq 2$, let $\mathcal{L}_{\leq h}(\mathcal{C}, \rho)$ be the set of these classes obtained when U has at most h layers of arbitrary unitary gates on disjoint supports of at most ρ physical subsystems. The depth bound applies to each complete circuit.

To prove finiteness, we extend the local rigidity argument of Eastin–Knill [2] and covariant error correction [9, 10] to bounded-depth circuit families, which need not form groups. For each support arrangement, we encode the gates, their inverses and the encoded action in a finitely generated complex coordinate algebra. The two-sided light-cone argument makes every base-field derivation annihilate the logical conjugation coordinates at every field-valued solution. These coordinates are constant modulo each minimal prime. Noetherianity then gives a finite evaluation image, even when the coordinate algebra is nonreduced. Conjugation determines a unitary up to phase, so this proves global finiteness without assuming a group structure on the circuit family.

Theorem 1. *Let $n, h \geq 1$, and $\rho \geq 2$ be integers, and let V encode a quantum code $\mathcal{C}$ satisfying Eq. (1) for a distance $d \geq 2$ on n physical subsystems. If*

$$h < 2\lceil \log_\rho d \rceil - 1, \tag{3}$$

then the logical actions of all circuits of depth at most h, with gates on at most ρ subsystems, that satisfy Eq. (2) obey

$$|\mathcal{L}_{\leq h}(\mathcal{C}, \rho)| < \infty.$$

All gate supports and unitary matrices are allowed. Consequently, $\mathcal{L}_{\leq h}(\mathcal{C}, \rho)$ is not dense in $\mathrm{PU}(K)$.

Proof. Equation (3) is equivalent to $\rho^{\lceil h/2 \rceil} < d$. Pad to h layers and n gate slots per layer, using identity gates on empty supports for unused slots. Empty-support scalar gates can be removed modulo global phase. There are

* Mailing address

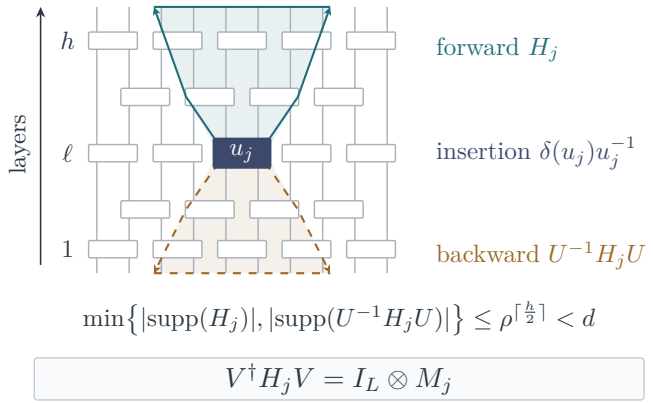


FIG. 1. Two-sided light cones. A derivation applied to gate u_j in layer ℓ gives the propagated insertion H_j. Under the depth condition, its forward support or the backward support of $U^{-1}H_jU$ is detectable. Error detection and the exact encoded action yield gauge-only compression of H_j in the output frame. The brickwall arrangement assumes no geometric locality.

finitely many ordered support arrangements α, so fix one of them. List its $J = nh$ slots in execution order, with $U = u_J \cdots u_1$.

For each slot introduce independent local matrix variables for the gate and its inverse; embed them in the full physical space as u_j, v_j. Introduce logical matrices W, W' and gauge matrices Z, Z'. Write $E = V^\dagger$, a constant complex matrix, and $Q = W \otimes Z$. The defining polynomial relations are

$$u_j v_j = v_j u_j = I, \qquad (4)$$
$$WW' = W'W = I, \quad ZZ' = Z'Z = I,$$
$$UV = VQ, \quad EU = QE. \qquad (5)$$

Let $R_\alpha = \mathbb{C}[\mathbf{x}]/I_\alpha$ be the resulting coordinate algebra. It is Noetherian because there are finitely many matrix-entry variables. Every physical circuit counted by the theorem gives a complex evaluation of R_α: use adjoints for the inverse variables. The second relation in Eq. (5) follows from unitarity and Eq. (2). General field-valued solutions need not be unitary; their inverse variables are not interpreted as adjoints.

Let $F/\mathbb{C}$ be any field extension, let $\phi : R_\alpha \to F$ be a complex-algebra homomorphism, and let $\delta : F \to F$ be any $\mathbb{C}$-linear derivation. We use the same symbols for evaluated matrices. Detection in Eq. (1) extends to F-valued supported operators by expanding them in the complex matrix-unit basis. Thus it remains a consequence of the original detection hypothesis. The product rule gives

$$H = \delta(U)U^{-1} = \sum_{j=1}^{J} H_j, \qquad H_j = U_{>j}\delta(u_j)u_j^{-1}U_{>j}^{-1}, \qquad (6)$$

where $U_{>j} = u_J \cdots u_{j+1}$. If slot j lies in layer ℓ, the forward support has at most $\rho^{h-\ell+1}$ sites. In the input frame,

$$U^{-1}H_jU = U_{<j}^{-1}u_j^{-1}\delta(u_j)U_{<j},$$

where $U_{<j} = u_{j-1} \cdots u_1$, and the backward support has at most ρ^ℓ sites. Other gates in layer ℓ have disjoint supports and cancel from these counts. Hence one of the supports has size at most $\rho^{\min\{\ell, h-\ell+1\}} \le \rho^{\lceil h/2 \rceil} < d$.

If the forward support is detectable, $EH_jV = I_L \otimes M_j$ directly. Otherwise detection gives $EU^{-1}H_jUV = I_L \otimes N_j$. Both encoding relations in Eq. (5) are needed to transfer this to the output frame:

$$\begin{aligned} EH_jV &= Q(EU^{-1}H_jUV)Q^{-1} \\ &= I_L \otimes ZN_jZ^{-1}. \end{aligned}$$

Thus, in a common frame,

$$EHV = I_L \otimes M. \qquad (7)$$

Differentiating $UV = VQ$, using $EV = I$ and $U^{-1}V = VQ^{-1}$, also gives

$$EHV = \delta(W)W^{-1} \otimes I_G + I_L \otimes \delta(Z)Z^{-1}. \qquad (8)$$

Comparing these equations at one diagonal gauge matrix entry ($g \ge 1$) shows that $\delta(W)W^{-1} = \lambda I_L$ for some $\lambda \in F$. Since $\delta(W^{-1}) = -W^{-1}\delta(W)W^{-1}$, every coordinate

$$r_{abce} = W_{ab}(W^{-1})_{ce} \qquad (9)$$

of $A \mapsto WAW^{-1}$ has zero δ-derivative.

This holds for every F, ϕ, δ. In particular it holds in the fraction field of $R_\alpha/\mathfrak{p}$ for each minimal prime $\mathfrak{p}$. Lemma 1 in the End Matter therefore gives finitely many joint evaluations of all r_{abce}. For unitary matrices, equal conjugation actions imply that their ratio commutes with every matrix and is a scalar of modulus one. Consequently each coordinate tuple represents at most one projective logical action. Taking the finite union over support arrangements proves finiteness. Since $K \ge 2$, the finite image cannot be dense in $\mathrm{PU}(K)$. □

Finiteness does not exclude a finite subset $S \subseteq \mathcal{L}_{\le h}(\mathcal{C}, \rho)$ for which

$$\overline{\langle S \rangle} = \mathrm{PU}(K).$$

The group $\langle S \rangle$ allows words of arbitrary finite length. A word of length r in $S \cup S^{-1}$ has a concatenated implementation of depth at most rh. These products are not required to remain in $\mathcal{L}_{\le h}(\mathcal{C}, \rho)$, so density of the generated group does not assert density within the original depth bound. For example, Dua et al. give a $[[37, 1, 7]]$ CSS code with separate depth-one circuits of gate range $\rho = 3$ that implement [11, Sec. V and Appendix B 3]

$$S = \{\overline{T}^\dagger, \sqrt{\overline{X}}^\dagger\} \subseteq \mathcal{L}_{\le 1}(\mathcal{C}, 3).$$

The two circuits use different partitions of the physical qubits. Since $\overline{H} \propto (\overline{T}^\dagger)^2 \sqrt{\overline{X}}^\dagger (\overline{T}^\dagger)^2$ and $\overline{T} = (\overline{T}^\dagger)^7$, this pair generates a dense subgroup of PU(2). For this code, $2\lceil \log_3 7 \rceil - 1 = 3$, so $\mathcal{L}_{\le 1}(\mathcal{C}, 3)$ is finite by Theorem 1.

In a fault-tolerance threshold construction based on the constant-depth unitary logical gadgets considered here, logical-gate compilation is necessary for universal control in the finite-image regime. On a fixed code, the available shallow logical actions form a finite alphabet. If its generated group is dense, arbitrary target logical unitaries can be approximated by words over this alphabet, with error correction interleaved as prescribed by the protocol [1, 11]. The constant-depth premise applies to each gadget. Compiled word lengths cannot remain uniformly bounded when arbitrary targets must be approximated to arbitrarily high accuracy.

The finiteness condition has an exact boundary for stabilizer codes. For an integer $\rho \ge 2$, define the critical depth

$$h_\infty(\mathcal{C}, \rho) = \min\{h \in \mathbb{Z}_{\ge 1} : \mathcal{L}_{\le h}(\mathcal{C}, \rho) \text{ is infinite}\}, \quad (10)$$

with value ∞ if the set is empty. When $d \ge 2$, Theorem 1 gives

$$h_\infty(\mathcal{C}, \rho) \ge 2\lceil \log_\rho d \rceil - 1. \quad (11)$$

For the upper bound of $h_\infty(\mathcal{C}, \rho)$, we implement continuous rotations generated by a weight-d logical Pauli. A Clifford tree collects its nonidentity factors onto at most ρ qubits in $\lceil \log_\rho d \rceil - 1$ layers. The compressed Pauli then fits in one allowed rotation gate. Conjugating back recovers the original Pauli rotation, at total depth $2\lceil \log_\rho d \rceil - 1$. This uses the established tree constructions for parity compression, phase gadgets, and Pauli-rotation synthesis [12–14]. The resulting continuous family is nonconstant modulo phase, giving the matching upper bound.

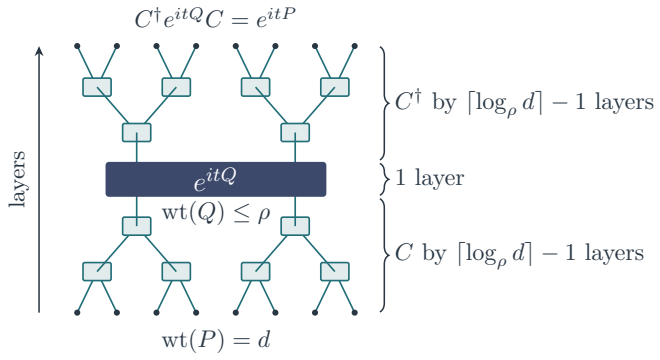


FIG. 2. Mechanism of Theorem 2. Time runs upward. C compresses the support of a weight-d logical Pauli to $Q = CPC^\dagger$, with $\mathrm{wt}(Q) \le \rho$. One rotation gate followed by $C^\dagger$ gives the complete logical circuit $C^\dagger e^{itQ} C = e^{itP}$, of depth $2\lceil \log_\rho d \rceil - 1$.

Theorem 2. *Let $\mathcal{C}$ be an $[[n, k, d]]$ qubit stabilizer code, with $k \ge 1$ and actual distance $d \ge 2$. Let $\rho \ge 2$ be an integer, and allow arbitrary unitary gates on any set of at most ρ qubits. Then*

$$h_\infty(\mathcal{C}, \rho) = 2\lceil \log_\rho d \rceil - 1. \quad (12)$$

At this depth, the logical image contains a nontrivial continuous one-parameter subgroup.

Proof. Put $m = \lceil \log_\rho d \rceil$. It suffices to construct infinitely many logical actions at depth at most $2m - 1$. Choose a Hermitian code-preserving Pauli P of weight d, with $P^2 = I$ and nonscalar logical action $\overline{P}$ [15, Sec. 3.2]. We first compress its support so that its rotation fits in one allowed gate.

We adapt the unitary parity-concentration part of the binary-tree in-line symmetry-verification circuit in Ref. [12, Sec. IV and Fig. 2(a)] to gates on at most ρ qubits. At each layer, partition the active factors into groups of ρ, with at most one smaller group. In each group, basis changes and CNOTs compose into one allowed Clifford gate collecting the product onto one qubit. The disjoint groups form one layer. If w_r is the weight after r layers, then $w_0 = d$ and $w_{r+1} = \lceil \frac{w_r}{\rho} \rceil$, giving $w_r = \lceil \frac{d}{\rho^r} \rceil$. Thus $m - 1$ layers yield C with $Q = CPC^\dagger$ of weight at most ρ. The compressed rotation e^{itQ} is one allowed gate. Conjugating back gives

$$U(t) = C^\dagger e^{itQ} C = e^{itP}. \quad (13)$$

The depth is at most $(m - 1) + 1 + (m - 1) = 2m - 1$. The complete circuit preserves the code and induces $e^{it\overline{P}}$. As $\overline{P}$ has both eigenvalues ± 1, its projective rotations form a nontrivial continuous U(1) subgroup of period π. Hence $h_\infty(\mathcal{C}, \rho) \le 2m - 1$, and the lower bound gives Eq. (12). □

The construction attains the threshold for an infinite exact logical image, but does not by itself provide fault tolerance against arbitrary single-gate faults. An error $e^{i\varepsilon Q}$ immediately after the central rotation gate propagates through $C^\dagger$ to $e^{i\varepsilon P}$, which induces a nontrivial logical error for $\varepsilon \notin \pi\mathbb{Z}$.

Conclusion. For general quantum codes in the finite-dimensional setting considered here, bounded-range circuits below the stated threshold realize only finitely many exact logical actions up to phase. The proof combines a detectable light cone in one of the two circuit directions with a finite-image criterion for the circuit coordinate algebra. Qubit stabilizer codes attain the bound by concentrating the support of a logical Pauli into one rotation gate and then reversing the compression. Together, the theorems locate the critical circuit depth to support infinitely many exact logical actions without imposing geometric locality.

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Lean proofs of the statements of Theorems 1 and 2, together with verification scripts, are available in the project repository (proof snapshot `ad6a378`). The proof of Theorem 1 follows the coordinate-algebra and derivation argument presented here, including scalar extension of detection, two-sided support bounds and the finite-evaluation-image criterion. The audited theorem dependencies contain no project axioms or `sorryAx`; their foundational axioms are `propext`, `Classical.choice` and `Quot.sound`. The quantitative bounds and computational discussion in the End Matter are additional mathematical deductions and are not formalized in this snapshot.

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END MATTER

Algebraic finite-image criterion

We give the global finiteness step in detail. The objects of interest are finitely many elements $r_1, \dots, r_m$ of a coordinate algebra R, representing the output coordinates of a polynomial map. An evaluation is a unital $\mathbb{C}$-algebra homomorphism $\phi : R \to \mathbb{C}$. We seek a finite set containing all output tuples $(\phi(r_1), \dots, \phi(r_m))$; neither the set of evaluations nor the whole parameter space is required to be finite.

For a minimal prime ideal $\mathfrak{p}$ of R, put

$$F_{\mathfrak{p}} = \mathrm{Frac}(R/\mathfrak{p}), \qquad \iota_{\mathfrak{p}} : R \longrightarrow R/\mathfrak{p} \longrightarrow F_{\mathfrak{p}}.$$

The quotient is an integral domain, so its map into its fraction field is injective and $\ker \iota_{\mathfrak{p}} = \mathfrak{p}$. A $\mathbb{C}$-linear derivation $\delta : F_{\mathfrak{p}} \to F_{\mathfrak{p}}$ is a linear map satisfying $\delta(xy) = \delta(x)y + x\delta(y)$. The derivations must be taken in these function fields: a $\mathbb{C}$-linear derivation of $\mathbb{C}$ itself is always zero and would give no constraint on the outputs.

Lemma 1. *Let R be a finitely generated commutative $\mathbb{C}$-algebra, and let $r_1, \dots, r_m \in R$. Suppose that, for every minimal prime $\mathfrak{p}$ and every $\mathbb{C}$-linear derivation $\delta : F_{\mathfrak{p}} \to F_{\mathfrak{p}}$,*

$$\delta\big(\iota_{\mathfrak{p}}(r_i)\big) = 0 \quad \textit{for every } i.$$

Then the joint evaluation image

$$\{(\phi(r_1), \dots, \phi(r_m)) : \phi : R \to \mathbb{C} \text{ a } \mathbb{C}\textit{-algebra map}\}$$

is finite.

Proof. If $R = 0$, there is no unital map $R \to \mathbb{C}$, and the image is empty. Assume $R \neq 0$.

Constants in a function field. We first explain why an element a of a field extension $F/\mathbb{C}$ that is killed by every $\mathbb{C}$-linear derivation must belong to $\mathbb{C}$. If a were transcendental, choose a transcendence basis B of $F/\mathbb{C}$ containing a. On the rational function field $L = \mathbb{C}(B)$, partial differentiation with respect to a defines a derivation with $\delta(a) = 1$, $\delta(b) = 0$ for $b \in B \setminus \{a\}$, and $\delta(c) = 0$ for $c \in \mathbb{C}$.

The extension F/L is algebraic and, in characteristic zero, separable. The derivation extends uniquely through this extension. To see the mechanism, let β have minimal polynomial $f(T) = T^s + \sum_{j=0}^{s-1} a_j T^j$ over L. Differentiating $f(\beta) = 0$ forces

$$\delta(\beta) = -\frac{\sum_{j=0}^{s-1} \delta(a_j)\beta^j}{f'(\beta)}.$$

Separability gives $f'(\beta) \neq 0$. For existence, extend δ from L to a derivation $L[T] \to L(\beta)$, using evaluation at β for the module structure and the displayed value for $\delta(T)$. The derivation kills f, and the Leibniz rule then makes it kill the whole ideal (f). It descends to $L[T]/(f) = L(\beta)$, proving existence as well as uniqueness. Extensions to finite intermediate fields agree on their overlaps, so they define a derivation on F. It still satisfies $\delta(a) = 1$, contradicting the assumed vanishing. Thus a is algebraic over $\mathbb{C}$, and algebraic closedness gives $a \in \mathbb{C}$.

Apply this argument to $F_{\mathfrak{p}}$. For each $i, \mathfrak{p}$, there is a constant $c_{i,\mathfrak{p}} \in \mathbb{C}$ with $\iota_{\mathfrak{p}}(r_i) = c_{i,\mathfrak{p}}$. The kernel calculation above therefore gives

$$r_i - c_{i,\mathfrak{p}} \in \mathfrak{p}. \tag{14}$$

The same minimal prime supplies a whole constant output tuple. At this stage we have a conclusion on each irreducible component; we still need a finite number of components and control of every evaluation, including points where components meet.

One polynomial for all evaluations. Hilbert's basis theorem makes R Noetherian. Hence its set of minimal primes is finite and nonempty [16]; write it as $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_t\}$. For each output coordinate form the monic polynomial

$$P_i(T) = \prod_{\nu=1}^{t} (T - c_{i,\mathfrak{p}_\nu}).$$

For any ν, one factor of $P_i(r_i)$ lies in $\mathfrak{p}_\nu$ by Eq. (14). Consequently

$$P_i(r_i) \in \bigcap_{\nu=1}^{t} \mathfrak{p}_\nu = \sqrt{(0)}.$$

The equality is the standard description of the nilradical as the intersection of the minimal primes [16]. By definition of the nilradical, some positive integer a_i satisfies $P_i(r_i)^{a_i} = 0$. Thus r_i satisfies the monic polynomial $P_i(T)^{a_i}$ in R; this is exactly the integrality statement used in the formal proof. We have not assumed that R is reduced.

Now let $\phi : R \to \mathbb{C}$ be *any* evaluation. Because ϕ preserves the complex coefficients,

$$0 = \phi\big(P_i(r_i)^{a_i}\big) = P_i\big(\phi(r_i)\big)^{a_i}.$$

A field has no nonzero nilpotents, so $P_i(\phi(r_i)) = 0$. In particular, the same polynomial P_i, chosen before ϕ, constrains every value of the coordinate r_i. Its root set $Z(P_i) \subset \mathbb{C}$ is finite since P_i is nonzero. The joint evaluation image is contained in the finite product

$$Z(P_1) \times \cdots \times Z(P_m),$$

which proves the lemma. □

Geometrically, the minimal primes describe the irreducible components of the affine parameter space. The field-derivation calculation makes each output coordinate constant on every such component, and Noetherianity limits their number. The polynomial argument also handles nilpotent directions: for example, in $R = \mathbb{C}[t, \epsilon]/(\epsilon^2)$, the element $r = \epsilon$ is nonzero but every complex evaluation sends it to zero. The parameter t remains arbitrary. Thus a finite output image does not imply that the coordinate algebra is reduced or that its spectrum is zero-dimensional. No smoothness assumption or restriction to regular points enters the argument.

For Theorem 1, take $R = R_\alpha$ and the K^4 coordinates in Eq. (9). The circuit calculation proves derivative vanishing for *every* field-valued solution, so in particular it applies to each canonical map $\iota_{\mathfrak{p}}$. The lemma therefore gives a finite set containing the outputs of all physical evaluations, including points on singular loci. At a physical point the coordinates give $A \mapsto WAW^\dagger$; two such maps agree exactly when the two unitaries differ by a scalar phase. Restricting the finite output image to physical evaluations therefore proves finiteness of the required projective logical actions for that support arrangement. This is the algebraic route implemented in the cited Lean snapshot.

Quantitative parameters and exact counting

The same component constants yield a cardinality bound. The kernel of any evaluation $R_\alpha \to \mathbb{C}$ contains a minimal prime $\mathfrak{p}$. All its logical conjugation coordinates therefore have the single joint value $(c_{i,\mathfrak{p}})_i$. Hence

$$|\mathcal{L}_{\le h}(\mathcal{C}, \rho)| \le \sum_{\alpha} \#\,\mathrm{Min}(R_\alpha). \tag{15}$$

Different components can give the same logical action, and some algebraic solutions need not be physical, so equality is not asserted.

Here are explicit parameters for the presentation in the proof. Write $q_i = \dim \mathcal{H}_i$, $P = \prod_i q_i$, and $D_j = \prod_{i \in S_j} q_i$, taking $D_j = 1$ on empty supports. With $J = nh$ slots, the number of complex variables is

$$N_\alpha = 2 \sum_{j=1}^{J} D_j^2 + 2K^2 + 2g^2. \tag{16}$$

There are at most $2JP^2 + 2K^2 + 2g^2 + 2PKg$ scalar defining relations, each of degree at most $\Delta = \max\{2, J\}$. The term $2JP^2$ counts the two inverse relations for each embedded physical gate; it is not an optimized presentation. The affine Bézout bound, counting irreducible components of all dimensions, gives $\#\,\mathrm{Min}(R_\alpha) \le \Delta^{N_\alpha}$ [17]. Nilpotents do not add components or complex evaluations.

A support arrangement assigns each site in each layer either to one of the n labelled slots or to no slot. There

are therefore at most $(n+1)^{nh}$ arrangements; disjointness and the arity restriction only reduce this count. Set $q_{\max} = \max_i q_i$ and $s = \min\{\rho, n\}$. Since $D_j \le q_{\max}^s$, under the hypotheses of Theorem 1 we obtain

$$|\mathcal{L}_{\le h}(\mathcal{C}, \rho)| \le (n+1)^{nh} \Delta^{N_*},$$
$$N_* = 2nh\, q_{\max}^{2s} + 2K^2 + 2g^2, \qquad \Delta = \max\{2, nh\}. \tag{17}$$

This bound is uniform over encodings with these dimensions, but very loose. For $n = 5$, $h = \rho = 2$, $q_{\max} = K = 2$, and $g = 1$, it is $6^{10}10^{330}$. A bound depending only on d, ρ, h cannot hold across all codes: the k-fold tensor power of the $[[5, 1, 3]]$ stabilizer code keeps $d = 3, \rho = 2, h = 1$, while its depth-one logical image contains at least 4^k projective logical Paulis.

Exact counting requires more than a component bound. If the encoding entries are supplied as exact algebraic numbers, introduce variables y_i for the logical conjugation coordinates and eliminate all original matrix-entry variables from $I_\alpha + (y_i - r_i)_i$. The resulting finite complex output variety gives candidate actions: its image is already finite and thus Zariski closed. For each candidate, retain its conjugation-coordinate equalities and test the original unitary and encoding equations in real and imaginary parts for solvability. Also, duplicate actions from different components and arrangements must be identified. Real quantifier elimination provides a decision procedure for this feasibility step [18]. The inverse variables of the enlarged complex model alone do not enforce physical unitarity.

These procedures are effective in principle with exact algebraic input, but the variable count and elimination costs make direct enumeration impractical in general. Arbitrary unspecified complex encoding coefficients do not supply an effective input representation, even though the theorem and Eq. (17) still hold. Exploiting symmetries, smaller presentations or a particular code may improve the calculation. Equations (15)–(17), the Bézout estimate and the exact-counting procedure are not formalized or executed by the cited Lean development.